An Equivalence Between Conservative Stability and Rational Expectations

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Abstract

We study two approaches to coalitional stability in infinitely repeated games. Greenberg's (1989) *Conservative Stable Standard of Behavior* (CSSB) is a set-valued solution concept that models worst-case reasoning: a coalition blocks only when it is guaranteed to gain across every continuation that the standard deems possible. *Perfect Coalitional Equilibrium* (PCE), introduced by Ali and Liu (2026), instead embodies rational expectations: a plan is self-enforcing if no coalition profitably blocks given the continuation play that players anticipate. We show that the two notions coincide. In Greenberg's coalitional repeated game situation, the set of PCE paths is the unique maximal nondiscriminating CSSB. Equivalently, every nondiscriminating CSSB consists of PCE paths, and the set of PCE paths is itself a nondiscriminating CSSB. Along the way, we establish two results of independent interest: the set of PCE paths is compact, and PCE admits an optimal penal code in the sense of Abreu (1988).

Keywords: repeated games; coalitions; core; stable standard of behavior; social situations; optimal penal codes.
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1 Introduction

A central insight of repeated game theory is that repeated interaction can enable self-interested players to sustain cooperation: the promise of future rewards and the threat of future punishments can deter deviations that are tempting in the short run. The classical theory develops this logic for individual deviations. In many settings, however, the relevant deviations are collective. Firms may jointly abandon a cartel, voters may form blocs, and workers and firms may rematch in pairs. When can the prospect of future punishment deter an entire coalition from blocking?

Answering this question requires specifying how a coalition evaluates the future consequences of its current deviation. One approach is based on conservatism, or worst-case thinking: a coalition blocks only if it is guaranteed to benefit, regardless of which continuation still regarded as possible ultimately occurs. This is the logic of Greenberg's (1989) *Conservative Stable Standard of Behavior*. A second approach invokes rational expectations: players share a common, self-consistent forecast of how play will unfold, and a coalition blocks only if blocking is profitable given that forecast. This logic underlies *Perfect Coalitional Equilibrium* (PCE), which we propose in Ali and Liu (2026). The two approaches thus embody sharply different views of coalitional beliefs off the equilibrium path. Does this difference lead to different predictions? For repeated strategic-form games, this note shows that it does not: the two approaches select exactly the same paths of play.

Conservative Stability. To state the result, we start by recalling Greenberg's conservative stability notion. Greenberg (1990)'s theory of social situations proposes a framework that describes strategic interactions through three ingredients: a set of *positions*—for a repeated game, the possible histories of play; an *inducement correspondence* that says, at each position, which positions a coalition can bring about by deviating; and the players' payoffs. A *standard of behavior* then prescribes, at every position, a set of continuation paths regarded as stable there. The standard is a Conservative Stable Standard of Behavior (CSSB) if it consists of exactly the paths that survive worst-case blocking: no prescribed path can be conservatively blocked, and every excluded path can be. A standard is *nondiscriminating* if it prescribes the same set of paths at every position. As a benchmark, Greenberg shows that if only individuals are allowed to deviate—i.e. in the *repeated game Nash situation*—conservative stability coincides with the notion of subgame perfect Nash equilibrium.

Theorem 0. (Greenberg 1989, Theorem 6.2) Let PEP denote the collection of equilibrium paths of all subgame perfect Nash equilibria (SPNE), and let σ^P be the standard of behavior (SB) defined by $\sigma^P(G) = PEP$ for every position G of the repeated game. Then σ^P is the unique maximal nondiscriminating CSSB for the repeated game Nash situation.

Theorem 0 shows that when only individuals are allowed to deviate, the maximal stable standard of behavior supported by worst-case reasoning coincides precisely with the set of SPNE paths—the paths consistent with rational expectations. Greenberg then asks whether this equivalence can be extended to coalitional deviations: what characterizes the set of CSSB in *coalitional repeated-game situations*? Although Greenberg (1989, Example 6.3, p. 292) obtains a sharp characterization for a particular common-interest game—the unique nondiscriminating CSSB selects the game's unique Pareto-optimal action profile—he stops short of a general characterization. In fact, writing later in his monograph (Greenberg 1990, Ch. 9), he is explicit that no general results have as yet been established.

Perfect Coalitional Equilibrium. We provide such a general characterization. To state it, we first recall the definition of PCE, introduced in Ali and Liu (2026). In a repeated strategic-form game, a plan specifies, after every history, the action profile to be played next. A plan is a PCE if, after every history, no coalition can profitably deviate from the prescribed action profile, where any deviation is evaluated using the continuation prescribed by the plan after the resulting history.

The key distinction between PCE and CSSB lies in their treatment of post-deviation play. PCE evaluates a deviation under the common continuation prescribed by the plan. CSSB, by contrast, evaluates the deviation against the worst case over a set of possible continuations and deems it profitable only if every coalition member benefits under every continuation in that set. Let $PCEP$ denote the set of paths induced by PCEs. With this notation, our main result is the exact coalitional analogue of Theorem 0.

Theorem 1. Let $PCEP$ denote the collection of equilibrium paths in all PCEs, and let σ^{PC} be the SB defined by $\sigma^{PC}(G) = PCEP$ for every position G of the repeated game. Then σ^{PC} is the unique maximal nondiscriminating CSSB for the coalitional repeated game situation.

The proof relies on two results about PCE, neither of which appears in Ali and Liu (2026) and both of which may be useful beyond the present paper. First, the set of PCE paths is compact in the product topology (Proposition 3). Second, PCE admits an optimal penal code in the sense of Abreu (1988): for each PCE path, a single profile of player-specific punishment paths deters every coalitional deviation (Proposition 4). Both results adapt the self-generation logic developed by Abreu (1988) and Abreu, Pearce, and Stacchetti (1990), with coalitions replacing individual players as the relevant deviators. On the CSSB side, we extend Propositions 5.3 and 5.4 in Greenberg (1989) from individual to coalitional deviations (Propositions 1 and 2).

Related Literature. Greenberg (1990) develops the theory of social situations as an integrative framework for cooperative and noncooperative models. In dynamic environments, the framework treats play as a process of sequential negotiation: a player or coalition may reject a proposed path by deviating, while a standard of behavior specifies the continuation paths regarded as possible from every resulting position. The central modeling choice is how a deviating coalition evaluates this set of continuations. Under the optimistic criterion, a deviation is profitable if at least one continuation admitted by the standard benefits every coalition member; under the conservative criterion, it is profitable only if every admitted continuation benefits every member.

The repeated-games literature develops both branches of this framework. On the conservative side, Greenberg (1989) analyzes nondiscriminating CSSB in repeated strategic-form games with only individual deviations, while Xue (2002) develops a distinct extension of Greenberg's conservative-stability logic to multistage coalitional deviations. On the optimistic side, Tadelis (1996) studies nondiscriminating OSSB in repeated extensive-form games. Greenberg, Monderer, and Shitovitz (1996) place repeated games within the broader class of multistage situations and extend the connection between conservative stability and subgame perfection.

The theory of social situations has also become an organizing language for farsighted coalitional stability. Xue (1998) uses standards of behavior to model coalitional stability under perfect foresight, and Mariotti and Xue (2003) develop this connection systematically. While stable standards of behaviors under optimistic expectations are closely related to farsighted stable sets, the conservative approach is consistent with Chwe (1994)'s largest consistent set; see Ray and Vohra (2015) for a review.

Our result identifies a canonical dynamic setting in which the distinction between conservatism and rational expectations disappears. While CSSB evaluates the deviation against the worst case over a set of possible continuations, PCE (Ali and Liu 2026) instead evaluates a deviation under the continuation prescribed by a common, self-enforcing plan. We show that, in repeated strategic-form games, the unique maximal nondiscriminating CSSB nevertheless selects exactly the PCE paths. Thus, despite their different treatments of off-path play, Greenberg's conservatism and rational expectations have the same behavioral implications in repeated strategic-form games with coalitional deviations. Methodologically, our intermediate results adapt the self-generation techniques of Abreu, Pearce, and Stacchetti (1990) and the optimal-penal-code logic of Abreu (1988) to coalitional blocking.

Outline. Section 2 recall Greenberg's repeated game situations and define the relevant notations; Section 3 defines PCE and states our main equivalent result; Section 4 states the four intermediate results behind Theorem 1. The proof of Theorem 1 and the proofs of the intermediate results are collected in the Appendix. Throughout, we stay as close as possible to Greenberg's notation, departing from it only when doing so improves exposition.

2 Model and Definitions

We first define the model using Greenberg's notation. Throughout, we maintain the following assumptions. Each player i's action set Z^i is a compact metric space, so that the set of action profiles $Z = \times_{i\in N} Z^i$ and—by Tychonoff's theorem—the path space $X = Z^\infty$ are compact in the product topology. Each stage payoff function $u_i : Z \to \mathbb{R}$ is continuous and bounded, and the players share a common discount factor $\delta \in [0, 1)$. Under these assumptions, each discounted payoff function $U_i : X \to \mathbb{R}$ defined below is continuous in the product topology and bounded; we use these facts repeatedly, for instance in the proofs of Proposition 1 and Proposition 3.

Let N be a finite set of players, $Z = \times_{i\in N} Z^i$ be the set of stage-game action profiles, $\{u_i\}_{i\in N}$ be the stage game payoff functions, and $X = Z^\infty$ denote the set of feasible paths in the repeated game. A repeated game Nash situation is a pair (γ, Γ), where Γ is the set of *positions* describing continuation situations (e.g. a subgame or history), and γ is the *inducement correspondence*: for each coalition $C \subseteq N$, position $G \in \Gamma$ and continuation path $x \in X$, $\gamma(C|G, x)$ denotes the positions coalition C can induce from the position G when the continuation path x is offered.

Formally, let Z^t denote the set of t-tuples of action profiles, then the set of positions $\Gamma := \cup_{t=0}^{\infty} Z^t$ is the set of histories. For each position $G = (z_1, z_2, \ldots, z_t) \in \Gamma$, define $a_i(G) = (1-\delta)\sum_{\tau=1}^{t} \delta^{\tau-1} u_i(z_\tau)$ and $b(G) = \delta^t$. Let $U_i : X \to \mathbb{R}$ denote player i's discounted payoff function given by $U_i(x) = (1-\delta)\sum_{\tau=1}^{\infty} \delta^{\tau-1} u_i(x_\tau)$ for all $x = (x_1, x_2, \ldots) \in X = Z^\infty$. The set of positions Γ then satisfies that for every $G \in \Gamma$,

$$N(G) = N; \quad X(G) = X; \text{ and } u_i(G) = a_i(G) + b(G)U_i \text{ for all } i \in N.$$

In the expressions above, $N(G)$ is the set of players at position G, $X(G)$ is the set of feasible outcomes at position G, and $u_i(G) : X \to \mathbb{R}$ is player i's payoff function over outcomes at position G. In a repeated game Nash situation, the set of players and feasible outcomes are constant across all positions; in Greenberg's language, player's payoff from continuation path $x \in X$ at position $G = (z_1, z_2, \ldots, z_t)$ is given by $u_i(G)(x) = a_i(G) + b(G)U_i(x)$; we will follow this notation in this note.

Conservative Stable Standard of Behavior. A *standard of behavior* (SB) is a mapping $\sigma : \Gamma \to 2^X$ that assigns to each position G a set of continuation paths $\sigma(G) \subseteq X$. Given an SB σ and a position G, a path $x \in X$ is said to be *conservatively dominated* at G (relative to σ) if there exists a coalition $C \subseteq N$ and $H \in \gamma(C|G, x)$ such that $\sigma(H) \neq \varnothing$ and $u_i(H)(y) > u_i(G)(x)$ for all $i \in C$ and $y \in \sigma(H)$.

Given an SB σ and a position G, the *conservative dominion* of G relative to σ, denoted $CDOM(\sigma, G)$, is the set of paths that are conservatively dominated at position G (relative to σ). An SB σ is:

- *conservatively internally stable* if $\sigma(G) \cap CDOM(\sigma, G) = \varnothing$ for all $G \in \Gamma$;
- *conservatively externally stable* if $X \backslash \sigma(G) \subseteq CDOM(\sigma, G)$ for all $G \in \Gamma$.

A Conservative Stable Standard of Behavior, or CSSB, is an SB that is both conservatively internally and externally stable. An SB σ is *nondiscriminating* if $\sigma(G) = \sigma(H)$ for all $G, H \in \Gamma$.

Remark 1 (Translating to standard repeated-games notation)**.** For readers more familiar with repeated games than with the theory of social situations, a position $G = (z_1, \ldots, z_t)$ is simply a t-period history, and $\Gamma = \cup_t Z^t$ is the set of all histories. The inducement correspondence γ records the histories reachable by an individual (respectively coalitional) one-shot deviation. The payoff decomposition $u_i(G)(x) = a_i(G) + b(G)U_i(x)$ separates the realized flow payoff $a_i(G)$ accrued along the history G

from the discounted continuation payoff $b(G)U_i(x) = \delta^t U_i(x)$ earned along the path x that follows. A nondiscriminating SB—one with $\sigma(G) = \sigma(H)$ for all G, H—is thus a history-independent prescription of continuation paths.

Repeated Game Nash Situation. In a repeated game Nash situation, the inducement correspondence $\gamma = \gamma^N$ allows only individual deviations and coalitions are not allowed to form. For each positive integer τ, position $G \in \Gamma$, path $x \in X$, player $i \in N$, and player i's action in period τ, $\zeta_\tau^i \in Z^i$, let $H = (G|x; \zeta_\tau^i) \in \Gamma$ denote the position $H = (G, x_1, x_2, \ldots, x_{\tau-1}, \zeta_\tau)$, where $\zeta_\tau \in Z^N$ is the action profile satisfying $\zeta_\tau^j = x_\tau^j$ for all $j \neq i$. For every $G \in \Gamma$ and $x \in X$, the inducement correspondence satisfies

$$\gamma^N(\{i\}|G,x) = \left\{ \left(G|x;\zeta_\tau^i\right) | \tau \in \{1,2,\ldots\} \text{ and } \zeta_\tau^i \in Z^i \right\} \text{ for all } i \in N$$
$$\gamma^N(C|G,x) = \varnothing \quad \text{for all } C \subseteq N \text{ with } \quad |C| > 1.$$

Note that in the repeated game Nash situation, since the inducement correspondence γ^N allows only individual deviations, a path $x \in X$ is conservatively dominated if and only if there exists player $i \in N$ and $H \in \gamma^N(\{i\}|G,x)$ such that $\sigma(H) \neq \varnothing$ and $u_i(H)(y) > u_i(G)(x)$ for all $y \in \sigma(H)$.

Coalitional Repeated Game Situation. A *coalitional repeated game situation* is a pair $(\gamma^{\mathcal{C}}, \Gamma)$ where Γ is defined identically to the repeated game Nash situation in Section 2, but the correspondence $\gamma^{\mathcal{C}}$ allows not only individual but also coalitional deviations. In particular, for each positive integer τ, $G \in \Gamma$, $x \in X$, $C \subseteq N$, and $\zeta_\tau^C \in Z^C$, let $H = (G|x; \zeta_\tau^C) \in \Gamma$ denote the position $H = (G, x_1, x_2, \ldots, x_{\tau-1}, \zeta_\tau)$, where $\zeta_\tau \in Z^N$ is the action profile satisfying $\zeta_\tau^j = x_\tau^j$ for all $j \notin C$. The inducement correspondence $\gamma^{\mathcal{C}}$ is given by: for every $G \in \Gamma$, $x \in X$, and $C \subseteq N$,

$$\gamma^{\mathcal{C}}(C|G,x) = \left\{ (G|x;\zeta_\tau^C) : \tau \in \{1,2,\ldots\} \text{ and } \zeta_\tau^C \in Z^C \right\}.$$

Given an SB σ, a path $x \in X$ is conservatively dominated at $G \in \Gamma$ if there exists a coalition $C \subseteq N$ and $H \in \gamma^{\mathcal{C}}(C|G,x)$ such that $\sigma(H) \neq \varnothing$ and $u_i(H)(y) > u_i(G)(x)$ for all $i \in C$ and $y \in \sigma(H)$. That is, a deviation is profitable for coalition C if every member of C strictly prefers every continuation at $\sigma(H)$ to x. By substituting γ with

$\gamma^{\mathcal{C}}$, the conservative dominion of G relative to σ, $CDOM(\sigma, G)$, can now be written as

$$CDOM(\sigma, G) = \Big\{x \in X : \exists C \subseteq N, \tau \geq 1, \zeta_\tau^C \in Z^C \text{such that } H = (G|x; \zeta_\tau^C) \text{ satisfies} \\ \sigma(H) \neq \varnothing \text{ and } u_i(H)(y) > u_i(G)(x) \text{ for all } y \in \sigma(H) \text{ and } i \in C\Big\}.$$

3 Main Result

We specialize Ali and Liu (2026)'s notion of PCE to the repeated strategic-form game that underlies the situations above. This is the environment of Example 1 of Ali and Liu (2026): the alternatives are action profiles $z \in Z$, and a blocking coalition C can bring about any profile in the effectivity set $E_C(z) = \{z' \in Z : z'_j = z_j \text{ for all } j \notin C\}$, that is, any profile obtained by changing only its own members' actions.

A *plan* $f : \Gamma \to Z$ specifies, at every position G, a default action profile $f(G)$ to be played next if no coalition blocks. A plan induces from each position G a continuation path, and we write $U_i(G \mid f)$ for player i's discounted payoff along that path. If the profile played at G is z (whether the default $f(G)$ or the profile $z' \in E_C(f(G))$ resulting from a block by some coalition C), the successor position is (G, z), from which the plan continues.

Definition 1. *Coalition $C \subseteq N$* ***profitably blocks*** *the plan f at position G if there exists $z' \in E_C(f(G))$ such that, for every $i \in C$,*

$$(1-\delta)\, u_i(z') + \delta\, U_i\big((G, z') \,\big|\, f\big) \;>\; U_i(G \mid f).$$

Definition 2. *A plan f is a* ***perfect coalitional equilibrium*** *(PCE) if no coalition profitably blocks it at any position $G \in \Gamma$. A path $x \in X$ is a* PCE path *if it is the continuation path that some PCE induces from the initial position G^0, and we let $PCEP \subseteq X$ denote the set of all PCE paths.*

A PCE is self-enforcing: given the continuation play it prescribes, no coalition gains by blocking at any position, on- or off-path. Because the continuation of a PCE at every position is itself a PCE, the set $PCEP$ has a recursive, self-generating structure that we exploit in Propositions 3 and 4.

Assumption 1. A PCE exists; equivalently, $PCEP \neq \varnothing$.

We maintain Assumption 1 throughout, just as Greenberg assumes the set of perfect equilibrium paths nonempty for his Theorem 0. It holds for every $\delta \in [0,1)$ whenever the stage game has a core action profile, and can fail only when that core is empty and players are sufficiently myopic (Ali and Liu 2026). The assumption is exactly what makes the maximal nondiscriminating CSSB well-defined: if $PCEP = \varnothing$, then by Theorem 1's argument any nondiscriminating CSSB σ would satisfy $\sigma(G^0) \subseteq PCEP = \varnothing$, so σ assigns the empty set to every position—and the empty standard is not externally stable. Hence $(\gamma^{\mathcal{C}}, \Gamma)$ would admit no nondiscriminating CSSB at all.

Theorem 1. *Let $PCEP$ denote the collection of equilibrium paths in all PCEs, and let σ^{PC} be the SB defined by $\sigma^{PC}(G) = PCEP$ for every $G \in \Gamma$. Then σ^{PC} is the unique maximal nondiscriminating CSSB for $(\gamma^{\mathcal{C}}, \Gamma)$.*

Theorem 1 rests on four intermediate results, which we state in the section below. We prove them and assemble them into a proof of Theorem 1 in the Appendix.

4 Intermediate Results

Let G^0 be the position associated with the empty history, i.e. at the beginning of the game. The first result shows that the closure of a nondiscriminating CSSB in the coalitional repeated game situation is also a CSSB.

Proposition 1. *Let σ be a nondiscriminating conservatively internally stable SB for the coalitional repeated game situation $(\gamma^{\mathcal{C}}, \Gamma)$. For each $G \in \Gamma$, let $\widetilde{\sigma}(G)$ be the closure of $\sigma(G)$. Then, the SB $\widetilde{\sigma}$ is also a nondiscriminating conservatively internally stable SB for $(\gamma^{\mathcal{C}}, \Gamma)$. In addition, if $\sigma(G^0) \neq \varnothing$, then $\widetilde{\sigma}(G^0)$ is also nonempty and compact.*

Let $x, y \in X$, let τ be a positive integer, let $C \subseteq N$ be a coalition, and let $\zeta_\tau^C \in Z^C$. Define

$$\left(x; \zeta_\tau^C; y\right) := \left(x_1, x_2, \ldots, x_{\tau-1}, \zeta_\tau, y_1, y_2, \ldots\right),$$

where $\zeta_\tau^j = x_\tau^j$ for all $j \notin C$. This denotes the path obtained by following x through period $\tau - 1$, then having the players in C deviate in period τ to ζ_τ^C, and then following the path y thereafter. The next result provides a characterization of nondiscriminating CSSB in the spirit of an optimal penal code.

Proposition 2. *Let σ be a nondiscriminating SB for the coalitional repeated game situation $(\gamma^{\mathcal{C}}, \Gamma)$, and suppose that for every $i \in N$ the set* $\arg\min\{U_i(x) : x \in \sigma(G^0)\}$ *is nonempty. For each $i \in N$, choose*

$$z(i|\sigma) \in \arg\min\{U_i(x) : x \in \sigma(G^0)\}.$$

Then a path $y \notin CDOM(\sigma, G^0)$ if and only if for every nonempty coalition $C \subseteq N$, every period $\tau \geq 1$, and every $\zeta_\tau^C \in Z^C$, there exists some player $i \in C$ such that $U_i(y) \geq U_i\big(y; \zeta_\tau^C; z(i|\sigma)\big)$.

The next result turns to PCE and shows that the set of equilibrium paths is compact in the product topology over $X = Z^\infty$.

Proposition 3. *The set $PCEP$ is compact.*

The next and final intermediate result obtains an optimal penal code for PCE, analogous to Abreu (1988)'s finding for subgame perfect Nash equilibria.

Proposition 4. *A path $x[0] \in X$ belongs to $PCEP$ if and only if there exists a family of paths $\{x[i]\}_{i \in N} \subseteq X$ such that for all $k \in \{0\} \cup N$, all $\tau \geq 1$, all coalitions $C \subseteq N$, and all $\zeta_\tau^C \in Z^C$, there exists $j \in C$ such that $U_j(x[k]) \geq U_j(x[k]; \zeta_\tau^C; x[j])$.*

These four results combine as follows. Propositions 1 and 2 are the coalitional analogues of Greenberg's nondiscrimination lemmas: Proposition 1 lets us replace any nondiscriminating CSSB by its closure, and so work with compact standards, while Proposition 2 reduces conservative stability to a single per-player test—a path is undominated exactly when, against every coalitional deviation, some member is held to at least her payoff under the standard's worst path for her. Propositions 3 and 4 then show that $PCEP$ passes precisely this test: it is compact, so each player's worst PCE path exists, and those worst paths form an optimal penal code that deters every coalitional deviation from every PCE path. Matching the two characterizations—of nondiscriminating CSSB and of $PCEP$—yields Theorem 1. We carry this out in Section A, with the proofs of Propositions 1 to 4 in Section B.

A Proof of Theorem 1

Since $\sigma^{PC}(G) = PCEP$ for all $G \in \Gamma$, σ^{PC} is a nondiscriminating SB. We first prove that σ^{PC} is a CSSB by proving conservative internal and external stability.

For internal stability, fix $x[0] \in \sigma^{PC}(G^0) = PCEP$. Since $PCEP$ is nonempty by Assumption 1 and compact by Proposition 3, and each U_i is continuous in the product topology, for every $i \in N$ there exists $z(i|\sigma^{PC}) \in \arg\min\{U_i(x) : x \in PCEP\}$. By Proposition 4, there exists a family of paths $\{x[i]\}_{i \in N} \subseteq PCEP$ such that for all $\tau \geq 1$, all coalitions $C \subseteq N$, and all $\zeta_\tau^C \in Z^C$, there exists $j \in C$ such that $U_j(x[0]) \geq U_j(x[0]; \zeta_\tau^C; x[j])$. Since by definition $U_j(x[0]; \zeta_\tau^C; z(j|\sigma^{PC})) \leq U_j(x[0]; \zeta_\tau^C; x[j])$, we have

$$U_j(x[0]) \geq U_j(x[0]; \zeta_\tau^C; z(j|\sigma^{PC})). \tag{1}$$

By Proposition 2, this implies $x[0] \notin CDOM(\sigma^{PC}, G^0)$. Since $x[0] \in \sigma^{PC}(G^0)$ was arbitrary, σ^{PC} is conservatively internally stable.

For conservative external stability, fix $x[0] \in X \backslash \sigma^{PC}(G^0) = X \backslash PCEP$. We will prove that $x[0] \in CDOM(\sigma^{PC}, G^0)$. Suppose, toward a contradiction, that $x[0] \notin CDOM(\sigma^{PC}, G^0)$. Proposition 2 then implies that for every nonempty coalition $C \subseteq N$, every period $\tau \geq 1$, and every $\zeta_\tau^C \in Z^C$, there exists $j \in C$ such that

$$U_j(x[0]) \geq U_j\big(x[0]; \zeta_\tau^C; z(j|\sigma^{PC})\big). \tag{2}$$

Now define $x[i] := z(i|\sigma^{PC})$ for all $i \in N$, so $\{x[i]\}_{i \in N} \subseteq PCEP$. Using arguments similar to those leading to (1), this implies that for every $i \in N$, every nonempty coalition $C \subseteq N$, every period $\tau \geq 1$, and every $\zeta_\tau^C \in Z^C$, there exists $j \in C$ such that

$$U_j(x[i]) \geq U_j\big(x[i]; \zeta_\tau^C; z(j|\sigma^{PC})\big). \tag{3}$$

Combining (2) and (3) and observing $z(j|\sigma^{PC}) = x[j]$, this shows that for all $k \in \{0\} \cup N$, all nonempty coalitions $C \subseteq N$, all periods $\tau \geq 1$, and all $\zeta_\tau^C \in Z^C$, there exists $j \in C$ such that $U_j(x[k]) \geq U_j(x[k]; \zeta_\tau^C; x[j])$. By Proposition 4, it follows that $x[0] \in PCEP$, contradicting the choice of $x[0]$. Therefore, $x[0] \in CDOM(\sigma^{PC}, G^0)$ for every $x[0] \in X \backslash \sigma^{PC}(G^0)$. Hence σ^{PC} is conservatively externally stable. Therefore σ^{PC} is a nondiscriminating CSSB.

It remains to prove maximality and uniqueness. Let σ be any nondiscriminating CSSB for $(\gamma^{\mathcal{C}}, \Gamma)$. Since every CSSB is in particular conservatively internally stable, Proposition 1 implies that its closure $\widetilde{\sigma}$ is a nondiscriminating conservatively internally stable SB with $\widetilde{\sigma}(G^0)$ compact. Because $\widetilde{\sigma}(G^0)$ is compact and each U_i is continuous in the product topology, the set $\arg\min\{U_i(x) : x \in \widetilde{\sigma}(G^0)\}$ is nonempty, so for each

$i \in N$ we may choose

$$\widetilde{z}(i|\widetilde{\sigma}) \in \arg\min\{U_i(x) : x \in \widetilde{\sigma}(G^0)\}.$$

Since $\widetilde{\sigma}$ is conservatively internally stable, Proposition 2 implies that for every $x \in \widetilde{\sigma}(G^0)$, every nonempty coalition $C \subseteq N$, every $\tau \geq 1$, and every $\zeta_\tau^C \in Z^C$, there exists $j \in C$ such that

$$U_j(x) \geq U_j(x; \zeta_\tau^C; \widetilde{z}(j|\widetilde{\sigma})).$$

Fix any $x[0] \in \widetilde{\sigma}(G^0)$ and define $\widetilde{x}[i] := \widetilde{z}(i|\widetilde{\sigma})$ for all $i \in N$, then $\{\widetilde{x}[k]\}_{k \in \{0\} \cup N} \subseteq \widetilde{\sigma}(G^0)$. The preceding inequality shows that the hypothesis of Proposition 4 is satisfied, so $\widetilde{x}[0] \in PCEP = \sigma^{PC}(G^0)$. Since $\widetilde{x}[0] \in \widetilde{\sigma}(G^0)$ was arbitrary, we have $\widetilde{\sigma}(G^0) \subseteq \sigma^{PC}(G^0)$. Because $\sigma(G^0) \subseteq \widetilde{\sigma}(G^0)$, it follows that $\sigma(G^0) \subseteq \sigma^{PC}(G^0)$ as well. Since both σ and σ^{PC} are nondiscriminating, the same inclusion holds for every $G \in \Gamma$. Thus σ^{PC} contains every nondiscriminating CSSB, so σ^{PC} is maximal. Finally, if σ is any maximal nondiscriminating CSSB, then the same inclusions apply and maximality of σ would imply that $\sigma = \sigma^{PC}$. Therefore σ^{PC} is the unique maximal nondiscriminating CSSB. ∎

B Proofs of the Intermediate Results

B.1 Proof of Proposition 1

Since σ is nondiscriminating, $\sigma(G) = \sigma(H)$ for all $G, H \in \Gamma$; taking closures yields $\widetilde{\sigma}(G) = \widetilde{\sigma}(H)$, so $\widetilde{\sigma}$ is also nondiscriminating. If $\sigma(G^0) \neq \varnothing$, then $\widetilde{\sigma}(G^0)$ is nonempty, and since X is compact, it is compact as well.

It remains to show conservative internal stability. Fix $x \in \widetilde{\sigma}(G^0)$ and suppose, toward a contradiction, that $x \in CDOM(\widetilde{\sigma}, G^0)$. Then there exist a coalition $C \subseteq N$, a period $\tau \geq 1$, and $\zeta_\tau^C \in Z^C$ such that, letting $H = (G^0|x; \zeta_\tau^C)$, we have $\widetilde{\sigma}(H) \neq \varnothing$ and

$$u_i(H)(y) > u_i(G^0)(x) \quad \text{for all } y \in \widetilde{\sigma}(H) \text{ and } \ i \in C. \tag{4}$$

Since $x \in \widetilde{\sigma}(G^0)$, by the definition of closure, there exists a sequence $\{x[m]\}_{m=1}^\infty \subseteq \sigma(G^0)$ with $x[m] \to x$ in the product topology. For each m, define $H^m = (G^0 \,|\, x[m]; \zeta_\tau^C)$, then $H^m \in \gamma^{\mathcal{C}}(C|G^0, x[m])$ for all m. Since σ is conservatively internally stable and $x[m] \in \sigma(G^0)$, we have that for all $m \geq 1$, $x[m] \notin CDOM(\sigma, G^0)$. Therefore by the definition of $CDOM(\sigma, G^0)$, for each m, there exist $y[m] \in \sigma(H^m)$ and $i_m \in C$ such that

$$u_{i_m}(H^m)(y[m]) \leq u_{i_m}(G^0)(x[m]). \tag{5}$$

Since C is finite, by passing to a subsequence we may assume that $i_m = i^* \in C$ for all m. Because σ is nondiscriminating, $\sigma(H^m) = \sigma(H)$ for all m, so $y[m] \in \sigma(H) \subseteq \widetilde{\sigma}(H)$ for all m. Since $\widetilde{\sigma}(H)$ is compact, we may also assume (after passing to a further subsequence) that $y[m] \to y^*$ for some $y^* \in \widetilde{\sigma}(H)$. By continuity,

$$u_{i^*}(H^m)(y[m]) \to u_{i^*}(H)(y^*), \qquad u_{i^*}(G^0)(x[m]) \to u_{i^*}(G^0)(x).$$

Taking limits in (5) yields $u_{i^*}(H)(y^*) \leq u_{i^*}(G^0)(x)$, which contradicts (4) since $y^* \in \widetilde{\sigma}(H)$ and $i^* \in C$. Thus $x \notin CDOM(\widetilde{\sigma}, G^0)$. Since $x \in \widetilde{\sigma}(G^0)$ was arbitrary, $\widetilde{\sigma}$ is conservatively internally stable. ■

B.2 Proof of Proposition 2

Fix $x \in X$. By definition, $x \in CDOM(\sigma, G^0)$ if and only if there exist a coalition $C \subseteq N$, a period $\tau \geq 1$, and $\zeta_\tau^C \in Z^C$ such that, letting $H = (G^0|x; \zeta_\tau^C)$, we have $\sigma(H) \neq \varnothing$ and $u_j(H)(y) > u_j(G^0)(x)$ for all $y \in \sigma(H)$ and $j \in C$.

Since σ is nondiscriminating, $\sigma(H) = \sigma(G^0)$. Furthermore, since $z(i|\sigma) \in \arg\min\{U_i(x) : x \in \sigma(G^0)\}$, we have $x \in CDOM(\sigma, G^0)$ if and only if there exist a coalition $C \subseteq N$, a period $\tau \geq 1$, and $\zeta_\tau^C \in Z^C$ such that $u_j(H)\big(z(j|\sigma)\big) > u_j(G^0)(x)$ for all $j \in C$.

Negating this statement, we obtain that $x \notin CDOM(\sigma, G^0)$ if and only if for every coalition $C \subseteq N$, every period $\tau \geq 1$, and every $\zeta_\tau^C \in Z^C$, there exists some player $i \in C$ such that $u_i(H)\big(z(i|\sigma)\big) \leq u_i(G^0)(x)$, where $H = (G^0|x; \zeta_\tau^C)$.

By definition, $u_i(H)\big(z(i|\sigma)\big) = U_i(x; \zeta_\tau^C; z(i|\sigma))$ and $u_i(G^0)(x) = U_i(x)$. Substituting these identities yields that $x \notin CDOM(\sigma, G^0)$ if and only if for every coalition $C \subseteq N$, every period $\tau \geq 1$, and every $\zeta_\tau^C \in Z^C$, there exists some player $i \in C$ such that $U_i(x) \geq U_i(x; \zeta_\tau^C; z(i|\sigma))$. ■

B.3 Proof of Proposition 3

We make use of the self-generation arguments in Abreu, Pearce, and Stacchetti (1990). First, we define the self-generation operator on the space of continuation paths. Given a set of paths $Y \subseteq X$, we use $\Psi(Y)$ to denote the set of paths that can be enforced as on-path equilibrium behavior of a PCE using continuation paths in Y, and begin by proving a series of lemmas that are useful for establishing Proposition 3.

Definition 3. *For any set $Y \subseteq X$, define*

$$\Psi(Y) := \Big\{x \in X : \text{for every } C \subseteq N,\ \tau \geq 1,\ \text{and } \zeta_\tau^C \in Z^C,$$
$$\exists \big\{\underline{x}[i]\big\}_{i \in N} \subseteq Y \text{ such that } U_i(x) \geq U_i(x; \zeta_\tau^C; \underline{x}[i]) \text{ for some } i \in C\Big\}.$$

Lemma 1. *The operator Ψ is monotone: if $Y \subseteq Y'$, then $\Psi(Y) \subseteq \Psi(Y')$.*

Proof. Immediate from the definition. ■

Lemma 2. *If $Y \subseteq X$ is compact, then $\Psi(Y)$ is compact.*

Proof. Since $\Psi(Y) \subseteq X$ and X is compact by Tychonoff's Theorem, it suffices to show that $\Psi(Y)$ is closed. Let $(x[m])_{m=1}^{\infty} \subseteq \Psi(Y)$ be a sequence converging to $x \in X$ in the product topology. Take an arbitrary coalition $C \subseteq N$, arbitrary period $\tau \geq 1$, and arbitrary $\zeta_\tau^C \in Z^C$. For each m, because $x[m] \in \Psi(Y)$, there exists a family $\{\underline{x}[m,i]\}_{i \in N} \subseteq Y$ such that there exists some player $i \in C$ such that

$$U_i(x[m]) \geq U_i(x[m]; \zeta_\tau^C; \underline{x}[m,i]). \tag{6}$$

Since Y is compact and N is finite, after passing to a subsequence if necessary, we may assume that for every $i \in N$, $\underline{x}[m,i] \to \underline{x}[i]$ for some $\underline{x}[i] \in Y$.

We claim that the family $\{\underline{x}[i]\}_{i \in N}$ can be used as punishments to enforce $x \in \Psi(Y)$ against deviation $\zeta_\tau^C \in Z^C$ by coalition C in period τ. To see why, recall that by (6), for each m there exists $i_m \in C$ such that

$$U_{i_m}(x[m]) \geq U_{i_m}(x[m]; \zeta_\tau^C; \underline{x}[m, i_m]). \tag{7}$$

Since C is finite, by passing to a further subsequence we may assume that $i_m = i^*$ for all m, for some fixed $i^* \in C$. By continuity of payoffs and the convergences

$$x[m] \to x, \qquad \underline{x}[m, i^*] \to \underline{x}[i^*],$$

taking limits in (7) yields $U_{i^*}(x) \geq U_{i^*}(x; \zeta_\tau^C; \underline{x}[i^*])$. Since $i^* \in C$, this verifies the enforceability conditions for x against the coalition C and deviation ζ_τ^C in period τ.

Because C, τ, and ζ_τ were arbitrary, we conclude that $x \in \Psi(Y)$. Hence $\Psi(Y)$ is closed, and therefore compact. ■

Lemma 3. *If $Y \subseteq X$ satisfies $Y \subseteq \Psi(Y)$, then $Y \subseteq PCEP$.*

Proof. Fix $x \in Y$. Since $x \in \Psi(Y)$, for every coalition C, period τ, and deviation ζ_τ^C from x, there exists a family of paths $\{\underline{x}[i]\}_{i \in N} \subseteq Y$ such that at least one member of C is deterred by the corresponding continuation path $\underline{x}[i]$.

Construct a plan as follows. On path, play x. After any history corresponding to a coalition deviation ζ_τ^C by coalition C in period τ, choose one player $i \in C$ satisfying $U_i(x) \geq U_i(x; \zeta_\tau^C; \underline{x}[i])$, and continue from period $\tau + 1$ onward according to the path $\underline{x}[i]$. Proceed recursively at every history.

By construction, after every deviation at every history, there is some member of the deviating coalition who does not strictly gain. Hence no coalition profitably blocks at any history. Thus the constructed plan is a PCE whose equilibrium path is x, so $x \in PCEP$. ■

Lemma 4. *$PCEP \subseteq \Psi(PCEP)$.*

Proof. Suppose $x \in PCEP$ is the equilibrium path generated by the PCE f. By the definition of a PCE, for each coalition C, period τ, and deviation ζ_τ^C, there exists a family of paths $\{x[i]\}_{i \in N}$ generated from the continuation play of f such that some player $i^* \in C$ is deterred from this deviation; furthermore, since the continuation of a PCE at any history is again a PCE, $\{x[i]\}_{i \in N} \subseteq PCEP$, so $PCEP \subseteq \Psi(PCEP)$. ■

Proof of Proposition 3 Define recursively $X^0 := X$, $X^{k+1} := \Psi(X^k)$ for $k \geq 0$. By Lemma 1 Ψ is monotone, and since X is compact, by Lemma 2 the sets $\{X^k\}_{k=0}^\infty$ form a decreasing sequence of compact sets. Therefore

$$X^\infty := \bigcap_{k=0}^{\infty} X^k$$

is compact. To show that $PCEP$ is compact, we will show that $PCEP = X^\infty$.

By Lemma 4, $PCEP \subseteq \Psi(PCEP)$. Since Ψ is monotone and $PCEP \subseteq X = X^0$, we have $PCEP \subseteq \Psi(X^0) = X^1$. Iterating this argument yields $PCEP \subseteq X^k$ for all $k \geq 0$, so $PCEP \subseteq X^\infty$. It remains to show that $X^\infty \subseteq PCEP$. By Lemma 3, it suffices to prove that $X^\infty \subseteq \Psi(X^\infty)$.

The claim is trivially true if $X^\infty = \varnothing$. Suppose $X^\infty \neq \varnothing$, and fix an arbitrary $x \in X^\infty$ and an arbitrary coalition $C \subseteq N$, period $\tau \geq 1$, and deviation $\zeta_\tau^C \in Z^C$. Note that $x \in X^{k+1} = \Psi(X^k)$ for every $k \geq 0$. So for each k, there exists a family $\{\underline{x}[k,i]\}_{i \in N} \subseteq X^k$ satisfying the enforceability condition of $\Psi(X^k)$ for x.

Since X is compact, taking subsequences if necessary, each $\underline{x}[k,i]$ converges to some $\underline{x}[i] \in X$. Consider any player $i \in N$ and any $K \geq 0$. For all $k \geq K$, since the sequence $\{X^k\}_{k=0}^\infty$ is decreasing, it follows that $\underline{x}[k,i] \in X^k \subseteq X^K$. Because X^K is compact, it is closed, so taking limits yields $\underline{x}[i] \in X^K$. Since K was arbitrary, we conclude that

$$\underline{x}[i] \in \bigcap_{K=0}^{\infty} X^K = X^\infty \text{ for all } i \in N.$$

We will show that $\{\underline{x}[i]\}_{i \in N}$ can enforce $x \in X^\infty$.

Since N is finite, we can select a single subsequence $\{k_\ell\}_{\ell=1}^\infty$ along which $\underline{x}[k_\ell, i] \to \underline{x}[i]$ for all $i \in N$. By the enforceability of x in $\Psi(X^{k_\ell})$, for each ℓ there exists $i_\ell \in C$ such that

$$U_{i_\ell}(x) \geq U_{i_\ell}\big(x; \zeta_\tau^C; \underline{x}[k_\ell, i_\ell]\big).$$

Since C is finite, by passing to a further subsequence we may assume that $i_\ell = i^*$ for all ℓ, for some fixed $i^* \in C$. By continuity of payoffs and the convergence $\underline{x}[k_\ell, i^*] \to \underline{x}[i^*]$, taking limits yields $U_{i^*}(x) \geq U_{i^*}\big(x; \zeta_\tau^C; \underline{x}[i^*]\big)$, so x is enforceable against coalition C deviating in period τ and choosing ζ_τ using punishments $\{\underline{x}[i]\}_{i \in N} \subseteq X^\infty$. Since the choices of $x \in X$, $C \subseteq N$, $\tau \geq 1$, and $\zeta_\tau^C \in Z^C$ above were arbitrary, this verifies $X^\infty \subseteq \Psi(X^\infty)$, so $PCEP = X^\infty$ and therefore is compact. ■

B.4 Proof of Proposition 4

We first prove the "only if" direction. Take an arbitrary $x[0] \in PCEP$. Since $PCEP$ is compact and each U_i is continuous on X, for every player $i \in N$ we can select $x[i] \in \arg\min\{U_i(x) : x \in PCEP\}$.

Since $\{x[k]\}_{k=0}^n \subseteq PCEP \subseteq \Psi(PCEP)$, for every $k \in \{0\} \cup N$, every $\tau \geq 1$, every coalition $C \subseteq N$, and every $\zeta_\tau^C \in Z^C$, there exist some player $j \in C$ and some $y \in PCEP$ such that

$$U_j(x[k]) \geq U_j(x[k]; \zeta_\tau^C; y).$$

Because $x[j]$ minimizes player j's payoff over $PCEP$, $U_j(x[j]) \leq U_j(y)$. Hence replacing the continuation path y by $x[j]$ can only lower player j's continuation payoff, so

$$U_j(x[k]; \zeta_\tau^C; x[j]) \leq U_j(x[k]; \zeta_\tau^C; y).$$

Combining the two inequalities above yields $U_j(x[k]) \geq U_j(x[k]; \zeta_\tau^C; x[j])$. Since $j \in C$, this proves the "only if" direction.

We now prove the "if" direction. Suppose there exists a family of paths $\{x[i]\}_{i \in N} \subseteq X$ such that for all $k \in \{0\} \cup N$, all $\tau \geq 1$, all coalitions $C \subseteq N$, and all $\zeta_\tau^C \in Z^C$, there exists $j \in C$ with

$$U_j(x[k]) \geq U_j(x[k]; \zeta_\tau^C; x[j]). \tag{8}$$

We construct a plan f as follows. There are $n+1$ states, indexed by $\{0\} \cup N$. In state k, the prescribed path is $x[k]$. If in state k a coalition C deviates at period τ via ζ_τ^C, choose one player $i = i(k, \tau, C, \zeta_\tau^C) \in C$ satisfying (8), and from the next period onward switch to the path $x[i]$.

We claim that f is a PCE. Consider any history at which the continuation path is along $x[k]$, $k \in \{0\} \cup N$. If coalition C deviates by choosing some $\zeta^C \in Z^C$, then by construction the continuation path following the deviation is $x[j]$ for some $j \in C$ satisfying (8). Thus player j does not strictly gain and this deviation is not profitable. Because the same argument applies along every path $x[k]$, no coalition can profitably deviate after any history. Therefore f is a PCE. Its equilibrium path is $x[0]$, so $x[0] \in PCEP$. ■

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